

ASSIGNMENT 7

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Server running:

```
vaishnavi@ubuntu26:~$ cd Desktop
vaishnavi@ubuntu26:~/Desktop$ gcc server.c -o server -lpthread
vaishnavi@ubuntu26:~/Desktop$ gcc client.c -o client -lpthread
vaishnavi@ubuntu26:~/Desktop$ ./server
Server started on port 8080...
New client connected
Message: helloo
Message: hiii
New client connected
Message: good morning
Message: weather
New client connected
Message: sunny
New client connected
Message: it is very hot
Client disconnected
Message: computer
Message: assignment
Message: course
Message: learn
Message: coding
Message: data structures
Message: algorithms
Message: maths
Client disconnected
Message: formula
Message: help me please
Message: study
Client disconnected
Message: notebook
Message: many terminals
Client disconnected
█
```

Client 1

```
vaishnavi@ubuntu26:~/Desktop$ ./client
Connected to server...
helloo
hiii
good morning
weather
sunny
it is very hot
exit
disconnecting client from server....
vaishnavi@ubuntu26:~/Desktop$
```

Client 2

```
vaishnavi@ubuntu26:~/Desktop$ ./client
Connected to server...
good morning
weather
sunny
it is very hot
computer
assignment
course
learn
coding
data structures
algorithms
maths
formula
help me please
study
exit
disconnecting client from server....
vaishnavi@ubuntu26:~/Desktop$
```

Client 3

```
vaishnavi@ubuntu26:~/Desktop$ ./client
Connected to server...
sunny
it is very hot
computer
assignment
course
learn
coding
data structures
algorithms
maths
exit
disconnecting client from server....
vaishnavi@ubuntu26:~/Desktop$
```

Client 4

```
vaishnavi@ubuntu26:~/Desktop$ ./client
Connected to server...
it is very hot
computer
assignment
course
learn
coding
data structures
algorithms
maths
formula
help me please
study
notebook
many terminals
exit
disconnecting client from server....
vaishnavi@ubuntu26:~/Desktop$
```

Features:

- 1.If a client wants to exit then he can type exit to disconnect from the server.
- 2.Mutex lock is used so that only one thread accesses shared data at a time. This helps to avoid errors when multiple clients work simultaneously.